ELSAYED YOUNIS

Junior Spatial Data Engineer | GIS & Remote Sensing Specialist
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PROFESSIONAL SUMMARY

Geography & GIS graduate (2026) specializing in **Spatial Data Engineering** and field-to-cloud geospatial pipelines. Experienced in bridging field surveying and remote sensing with modern data engineering stacks—including **Python, SQL, PostgreSQL/PostGIS, Apache Spark/Sedona, Airflow, Kafka, dbt, and Azure Databricks**—to build scalable spatial data warehouses and automated ETL workflows.

TECHNICAL SKILLS

- **Spatial Data Engineering & Cloud:** PySpark, Apache Sedona, Airflow, Kafka, dbt, DuckDB, Azure Databricks, Azure Data Factory, GeoParquet, COG, Uber H3 Indexing.
- **Databases & Programming:** PostgreSQL/PostGIS, pgRouting, SQL Server, NoSQL | Python (GeoPandas, Rasterio, Shapely, ArcPy, WhiteboxTools, Pandas), SQL, Bash, HTML/CSS/JS.
- **GIS, Remote Sensing & BI:** ArcGIS Pro, ArcGIS Online, QGIS, ERDAS IMAGINE, FME Form, Google Earth Engine (GEE), MAPP Enterprise, Power BI (DAX), MicroStrategy.
- **Surveying, CAD & DevOps:** AutoCAD, Civil 3D, Total Station, Leveling | Linux (WSL2/Ubuntu), Docker, Git/GitHub.

PROFESSIONAL EXPERIENCE

Surveying & Construction Technician | Concord Engineering & Contracting *(New Alamein | May 2024 – Dec 2024)*

- Executed high-precision surveying (Total Station, Leveling) across an 8-month road infrastructure project, maintaining **±2 cm accuracy** over **500+ points**.
- Validated spatial layouts and resolved on-site coordinate discrepancies using **AutoCAD** and **Civil 3D**.

Surveying Specialist | El-Saeedy Contracting Office *(Menoufia | Mar 2022 – Nov 2023)*

- Led end-to-end field-to-office topographic surveys for road paving projects, generating elevation profiles and CAD deliverables from **1,000+ raw field points**.

INTERNSHIPS & INTENSIVE TRAINING

Microsoft Data Engineer Trainee | DEPI (MCIT & AMIT Learning) *(Jul 2026 – Dec 2026 | 6-Month Track)*

- Built scalable ETL/ELT pipelines, data warehouses, and big data workflows using **Python, SQL, and Microsoft Azure** (Data Factory, Databricks, Synapse), culminating in an end-to-end Capstone Project.

Data Analyst Specialist Trainee | DEPI (MCIT & EYouth) *(Jul 2025 – Dec 2025 | 6-Month Track)*

- Delivered **5+ analytics projects** and dimensional data models (Fact/Dimension), cutting data processing time by **40%** via automated ETL and **Power BI/SQL** dashboards (*Graduation Project: Bank Loan Performance Analysis*).

GIS & Remote Sensing Intern | Localia Geospatial Services & EDGE Pro *(Cairo | Jul 2025 – Aug 2025)*

- Processed Sentinel-2/drone imagery (**ERDAS IMAGINE**), produced **10+ thematic maps** (**ArcGIS Pro/QGIS**), and built interactive web GIS apps (**MAPP Enterprise, JS**).

Additional Short Internships:

- **Data Analysis Trainee** — National Telecommunication Institute (NTI) *(Apr 2026)*: EDA, Python, SQL & Power BI.
- **Remote Sensing Trainee** — NARSS *(Cairo | 1-Month Summer Internship)*: Radiometric correction & image classification.

SELECTED PROJECTS

- **Suez Canal Region Spatial & Hydrological Modeling (Graduation Project):** Automated Python, **GEE**, and **WhiteboxTools** workflows for DEM terrain analysis, flood hazard zoning, LST mapping, and SAR oil spill detection.
- **Lake Nasser Water Surface Monitoring (2015–2025):** Conducted a 10-year spatiotemporal shoreline and water volume change study using multi-temporal **Landsat-8/9** imagery and **NDWI** automation in ArcGIS Pro & ERDAS.

EDUCATION

B.A. in Geography & Geographic Information Systems (GIS) | Menoufia University *(Sep 2022 – Jun 2026)*

- Focus: Spatial Database Management, Remote Sensing, Spatial Analysis, and Cartography.

KEY CERTIFICATIONS

- **Data Engineering & Cloud:** ETL & Data Pipelines with Shell, Airflow & Kafka (**IBM**) | Advanced & Fundamentals of Azure Databricks (**Packt**) | Data Warehousing for BI (**Univ. of Colorado**) | Data Warehouse & NoSQL (**IBM**).
- **Geospatial Engineering & GIS:** Geospatial Data Engineering & Real-Time Pipelines Specialization (**Coursera**) | Spatial Data Science MOOC, Cluster Analysis & Climate Action MOOCs (**Esri**) | Spatial Modeling (**GeoSteps**).
- **Professional Development:** McKinsey Forward Program (**McKinsey & Company**).

LANGUAGES

Arabic: Native | **English:** B1 (Working Proficiency) | **German:** A2 (Elementary)